Tyler LaBonte

Senior Researcher
Microsoft Research AI Frontiers
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Research Interests

My research goal is to ensure **robust, grounded, and trustworthy multimodal agents** through foundational innovations in representation and generalization. Recently, I was a core member of the team behind Phi-4-reasoning-vision, a 15B multimodal vision-language reasoning model with application to computer-use agents. My PhD work focused on theory and algorithms for robustness to distribution shift.

Education

Georgia Institute of Technology — 2021–2026
Ph.D., Machine Learning
Minor in Mathematics
Advisors: Vidya Muthukumar and Jacob Abernethy

University of Southern California — 2017–2021
B.S., Applied and Computational Mathematics, *magna cum laude*
Minor in Computer Science
Advisor: Shaddin Dughmi

Employment

1. Microsoft Research AI Frontiers — Remote
   Senior Researcher — 2026–
   Future capabilities of multimodal agents.
2. Microsoft Research AI Frontiers — Redmond, WA
   Machine Learning Research Intern — 2025
   Advisors: Vibhav Vineet and Neel Joshi
   - Joined core development team of a 15 billion parameter multimodal vision-language model.
   - Led integration effort, tool-use training, and evaluation for a multimodal computer-use agent application.
3. Google — Sunnyvale, CA
   Machine Learning Research Intern — 2023
   Advisor: Kun Lin
   - Developed techniques to leverage Gemini LLM to improve architecture-agnostic hardware-software code design.
   - Synthesized chain-of-thought and few-shot prompting strategies to generalize to data-scarce applications.
4. Microsoft Research — Redmond, WA
   Machine Learning Research Intern — 2021–2022
   Advisors: Vibhav Vineet and Neel Joshi
   - Developed Transformer model for weakly supervised object detection via multiple instance learning.
   - Integrated pipeline into production system, enabling rapid delivery of new Windows Action Center capability.

5. GOOGLE X — Mountain View, CA
   Machine Learning Research Intern — 2020
   Advisor: Daniel R. Silva
   - Invented CNN-LSTM for temporal identity preservation in multiple object tracking for computational agriculture.
   - Presented results to Google executives, who approved an FTE hire to deploy my research to production systems.
6. SANDIA NATIONAL LABORATORIES — Albuquerque, NM
   Machine Learning Research Intern — 2019–2020
   Advisors: Carianne Martinez and Scott A. Roberts
   - Invented novel Bayesian CNN deep learning architecture which scales to billion-voxel 3D segmentation volumes.
   - Enabled error bound calculation for physical properties of graphite electrodes and thermal protection systems.

Publications

An asterisk (*) denotes equal contribution. An ($\alpha\beta$) denotes alphabetical author ordering.

PREPRINTS

1. SGD Provably Prioritizes a Shortcut Spurious Feature in the XOR Model
   Tyler LaBonte and Vidya Muthukumar
   Preprint, 2026

TECHNICAL REPORTS

1. Phi-4-reasoning-vision-15B Technical Report
   ($\alpha\beta$) Jyoti Aneja, Michael Harrison, Neel Joshi, Tyler LaBonte, John Langford, and Eduardo Salinas
   Technical Report, 2026
   Press coverage: Microsoft Research Blog, VentureBeat, Forbes
2. We Know Where We Don't Know: 3D Bayesian CNNs for Credible Geometric Uncertainty
   Tyler LaBonte, Carianne Martinez, and Scott A. Roberts
   Technical Report, 2019

CONFERENCE ARTICLES

1. Task Shift: From Classification to Regression in Overparameterized Linear Models
   Tyler LaBonte*, Kuo-Wei Lai*, and Vidya Muthukumar
   AISTATS 2025
   INFORMS Applied Probability Society Conference 2025
   IMS Workshop on Frontiers of Statistical Machine Learning 2025 **(top-10 award)**
2. The Group Robustness is in the Details: Revisiting Finetuning under Spurious Correlations
   Tyler LaBonte, John C. Hill, Xinchen Zhang, Vidya Muthukumar, and Abhishek Kumar
   NeurIPS 2024
3. Towards Last-layer Retraining for Group Robustness with Fewer Annotations
   Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
   NeurIPS 2023
4. Scaling Novel Object Detection with Weakly Supervised Detection Transformers
   Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
   WACV 2023

JOURNAL ARTICLES

1. On the Unreasonable Effectiveness of Last-layer Retraining
John C. Hill, Tyler LaBonte, Xinchen Zhang, and Vidya Muthukumar
Transactions on Machine Learning Research, 2026.
2. Student Misconceptions of Dynamic Programming: A Replication Study
Michael Shindler, Natalia Pinpin, Mia Markovic, Frederick Reiber, Jee Hoon Kim, Giles Pierre Nunez Carlos, Mine Dogucu, Mark Hong, Michael Luu, Brian Anderson, Aaron Cote, Matthew Ferland, Palak Jain, Tyler LaBonte, Leena Mathur, Ryan Moreno, and Ryan Sakuma.
Computer Science Education, 32(3):288–312, 2022
3. Quantifying the Unknown Impact of Segmentation Uncertainty on Image-Based Simulations
Michael C. Krygier, Tyler LaBonte, Carianne Martinez, Chance Norris, Krish Sharma, Lincoln N. Collins, Partha P. Mukherjee, and Scott A. Roberts
Nature Communications, 12(1):5414, 2021

WORKSHOP ARTICLES

1. On the Unreasonable Effectiveness of Last-layer Retraining
John C. Hill, Tyler LaBonte, Xinchen Zhang, and Vidya Muthukumar
ICLR 2025 Workshop on Spurious Correlations and Shortcut Learning
2. Saving a Split for Last-layer Retraining can Improve Group Robustness without Group Annotations
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
ICML 2023 Workshop on Spurious Correlations, Invariance, and Stability
3. Dropout Disagreement: A Recipe for Group Robustness with Fewer Annotations
Tyler LaBonte, Vidya Muthukumar, and Abhishek Kumar
NeurIPS 2022 Workshop on Distribution Shifts
4. Scaling Novel Object Detection with Weakly Supervised Detection Transformers
Tyler LaBonte, Yale Song, Xin Wang, Vibhav Vineet, and Neel Joshi
CVPR 2022 Workshop on Transformers in Vision

THESES

1. Generalization in Dynamic Environments: Foundations and Algorithms
Tyler LaBonte
Doctoral Dissertation, Georgia Institute of Technology, 2026
2. Finding the Needle in a High-Dimensional Haystack: Oracle Methods for Convex Optimization
Tyler LaBonte
Undergraduate Thesis, University of Southern California, 2021
Winner of the USC Discovery Scholar distinction

Awards

1. ICLR 2026 Top-200 Reviewer ($675) 2026
2. IMS Workshop on Frontiers of Statistical Machine Learning Travel Grant ($500) 2025
3. 2nd Place Research Talk/Poster Presentation – DOD NDSEG CONFERENCE 2023
4. Simons Institute Deep Learning Theory Workshop Travel Grant ($2,000) 2022
5. DoD National Defense Science and Engineering Graduate Fellowship ($170,000) 2021
    - One of two undergraduates to receive both DoD NDSEG and NSF GRFP in Computer Science

6. NSF Graduate Research Fellowship ($138,000—declined) 2021
7. USC Discovery Scholar (Research distinction for <100 USC graduates) 2021
8. Neo Scholar (Top ~100 CS undergraduates in America) – NEO 2020
9. U.S.S. Bowfin Memorial Scholarship ($5,000) 2020
10. 1st Place Computer Vision Project – TREEHACKS, STANFORD UNIVERSITY 2019
11. 1st Place Healthcare AI Project – TREEHACKS, STANFORD UNIVERSITY 2019
12. 1st Place Data Analytics Project – HACKSC, USC 2019
13. Admiral Bernard Clarey Memorial Scholarship ($7,000) 2018
14. National Top 20 Ethical Hacking Finalist – MAJOR LEAGUE HACKING 2018
15. USC Trustee Scholar ($250,000) 2017
16. USC Viterbi Fellow ($24,000) 2017
17. Dolphin Scholarship ($13,600) 2017
18. Rear Admiral Paul Lacy Memorial Scholarship ($6,500) 2017
19. National Merit Scholar ($3,000) 2017

Talks

1. University of Washington School of Computer Science and Engineering – SEATTLE, WA 2025
   Task Shift: From Classification to Regression in Overparameterized Linear Models
2. Georgia Tech School of Industrial and Systems Engineering – ATLANTA, GA 2024
   Task Shift: From Classification to Regression via Benign Overfitting
3. Georgia Tech Machine Learning Center – ATLANTA, GA 2024
   Task Shift: From Classification to Regression via Benign Overfitting
4. Google DeepMind – MOUNTAIN VIEW, CA 2023
   Towards Last-layer Retraining for Group Robustness with Fewer Annotations
5. Google Cloud Technical Infrastructure – SUNNYVALE, CA 2023
   Large Language Models for Hardware-Software Code Design
6. DoD NDSEG Conference – SAN ANTONIO, TX 2023
   Towards Last-layer Retraining for Group Robustness with Fewer Annotations
7. Microsoft Research – REDMOND, WA 2021
   Weakly Supervised Detection Transformers for Effortless Computer Vision
8. USC Computer Science Theory Group – LOS ANGELES, CA 2021
   The Distance Oracle for Convex Optimization
9. Google X – MOUNTAIN VIEW, CA 2020
   Temporal Identity Preservation in Multiple Object Tracking
10. USC Computer Science Theory Group – LOS ANGELES, CA 2019
    3D Bayesian CNNs for Credible Geometric Uncertainty
11. USC Center for Artificial Intelligence in Society – LOS ANGELES, CA 2019
    3D Bayesian CNNs for Credible Geometric Uncertainty

12. USC Center for Artificial Intelligence in Society – Los Angeles, CA — 2019
Machine Learning Fairness in Word Embeddings

Advising

1. Xinchen Zhang – Georgia Tech MS → Synovus — 2024–2025
2. John C. Hill – Georgia Tech BS/MS → Georgia Tech PhD — 2022–2024

Teaching

1. Lecturer/Teaching Assistant (8 lectures) | Georgia Institute of Technology — 2024
CS 7545: Machine Learning Theory
2. Lecturer/Teaching Assistant (12 lectures) | Georgia Institute of Technology — 2023
CS 7545: Machine Learning Theory
3. Undergraduate Teaching Assistant | University of Southern California — 2021
CSCI 270: Introduction to Algorithms and Theory of Computing
4. Instructor | USC Center for Artificial Intelligence in Society — 2019
Introduction to Machine Learning
5. Undergraduate Teaching Assistant | University of Southern California — 2018
CSCI 170: Discrete Methods in Computer Science

Academic Service

Reviewing

1. NeurIPS — 2023, 2024, 2026
2. CVPR — 2026
3. ICLR — 2024, 2026
4. TMLR — 2025, 2026
5. CVPR Workshop on Demographic Diversity in Computer Vision — 2025
6. ICLR Workshop on Spurious Correlations and Shortcut Learning — 2025
7. ICML — 2025

Organizing

1. Organizer, Georgia Tech ML Theory Reading Group — 2021–2023, 2025
2. System Administrator, Georgia Tech ML Theory GPU Cluster — 2022–2025
3. Student Organizer, Learning Theory Alliance Mentorship Workshop — 2023

Open Source Software

1. Milkshake: Quick and extendable experimentation with classification models — 2023
https://github.com/tmlabonte/milkshake — ★ 5 3

2. WS-DETR: Weakly supervised Transformers for scaling novel object detection 2021–2022
   https://github.com/tmlabonte/weakly-supervised-detr ★ 10 ⑂ 6
3. BCNN: 3D Bayesian CNNs for credible geometric uncertainty 2019–2020
   https://github.com/sandialabs/bcnn ★ 62 ⑂ 19
   Transitioned to a production environment by Sandia National Laboratories
   19th most starred Sandia repository out of 693 (March 2025)
4. Tendies: Decoupling deep learning development and deployment 2018
   https://github.com/tmlabonte/tendies ★ 37 ⑂ 11
   Transitioned to a production environment by the Air Force Research Laboratory

Other Activities

1. Fleet Captain, Georgia Tech Sailing Club 2023–2025
2. House Chair, USC Hawai'i Club 2020–2021
3. Vice President of Finance, USC Hawai'i Club 2019–2020